Causally Attributable Anthropogenic Fingerprints in Damodar Valley Corporation Reservoir Storage

B.Tech Final Year Project — First-Level Departmental Review

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Outline

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2. Study Area
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4. Results
5. Claim Verdicts
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Motivation & Problem

Study Area

Methodology

Results

Claim Verdicts

Limitations & Next Steps

Why Dam Operations Fool Satellites

The GRACE problem

India uses GRACE-FO TWS data for drought alerts and water policy (NITI Aayog, Jal Shakti Ministry). GRACE measures at ≈330 km resolution — but the Damodar basin is only ∼23,370 km^2, smaller than one satellite pixel.

When a dam releases water for flood control, GRACE records it as a *climate anomaly*. It cannot tell the difference between a drought and a dam doing its job.

The DVC case

The DVC operates Maithon, Panchet, and Durgapur in tandem — one fills while the other releases, every monsoon. This operational signal is invisible to coarse sensors but detectible in high-resolution radar imagery.

Scale mismatch

Damodar basin ($23{,}370\,\mathrm{km}^2$) fits inside one GRACE mascon ($\sim 90{,}000\,\mathrm{km}^2$). Basin-level attribution needs a different method.

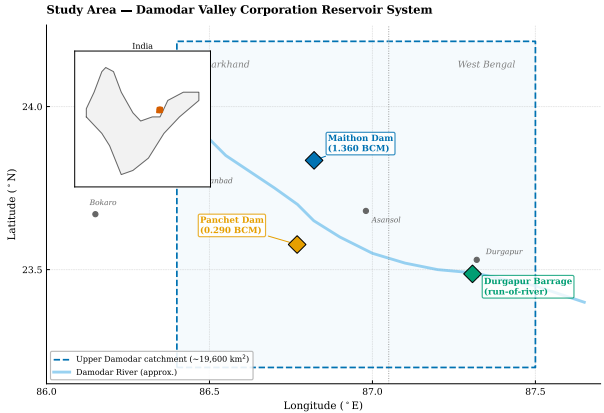
This study

Reconstruct reservoir storage month by month from Sentinel-1 SAR, separate the climate fraction with XGBoost, and test whether the residual is *causally* linked to dam operations.

Policy implication

Any drought index using raw GRACE data for the Damodar basin is systematically biased.

The DVC Cascade — Study Area



Map: DVC structures, reservoir outlines (Sentinel-1), and Damodar basin inset within India.

Structure	Cap.	Role
Maithon Dam	1.36 BCM	Flood, power
Panchet Dam	0.29 BCM	Tandem flood
Durgapur Barrage	RoR	Rabi irrigation

Why this basin is ideal:

- Same climate zone, different operating rules — built-in controlled contrast
- 2009 flood on record (Maiti & Bhattacharya 2011) — ground-truth validation
- Rabi drawdown visible from space every Oct–Mar

Period: 2017–2024 (Sentinel-1 era), $n = 82$ monthly observations per reservoir

Three Pre-Registered Claims

Pass/fail thresholds were written down *before* any analysis ran. No post-hoc threshold adjustments.

C1 — Attribution

DVC operations leave a causal, statistically attributable residual that a climate-only model cannot explain.

Pass: Granger or CCM test, $p < 0.05$ on STL-deseasonalised series

C2 — Fingerprint taxonomy

Each reservoir has a distinct operational signature classifiable from satellite data, with no ground sensors.

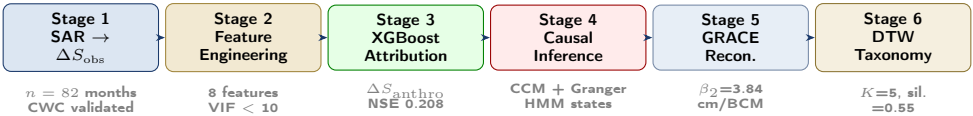
Pass: DTW silhouette score > 0.45, clusters agree with known events

C3 — GRACE contamination

The anthropogenic storage signal independently explains a fraction of GRACE-FO TWS variance after controlling for climate.

Pass: partial regression β_2, HAC-corrected $p < 0.05$

Six-Stage Analysis Pipeline



Satellite data

Sentinel-1 SAR
GRACE-FO RL06.3
MODIS ET/NDVI
ICESat-2

Climate drivers

CHIRPS rainfall
ERA5-Land (PET,SM)
Nino-3.4, DMI
6-month API

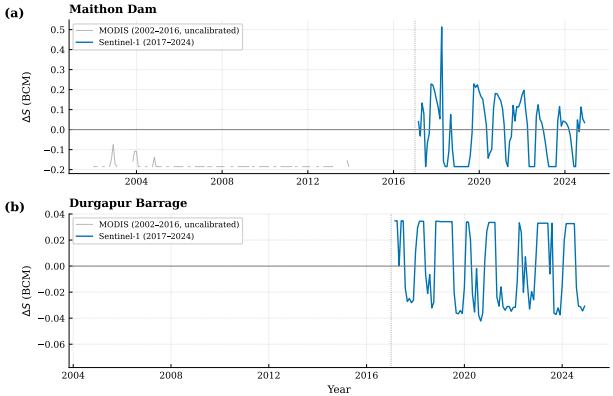
ML stack

XGBoost+Optuna
SHAP beeswarm
pyEDM (CCM)
hmmlearn, DTW

Validation

CWC bulletins
Expanding CV
999 surrogates
2009 flood event

Stage 1 — Storage Reconstruction



Monthly ΔS (BCM), 2017–2024. Shaded = 90 % bootstrap CI.

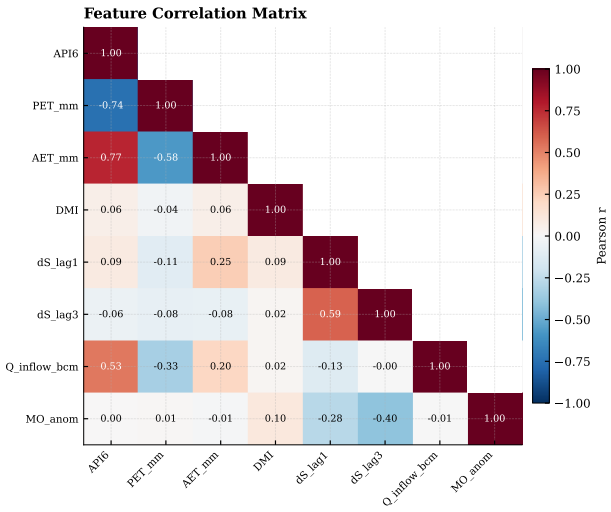
Sentinel-1 SAR penetrates monsoon cloud — the most operationally critical months for DVC. JRC water mask prevents rice paddy misclassification. CartoDEM A–V curves calibrated with ICESat-2.

CWC validation ($n = 9$)

RMSE (all points)	0.23 m
RMSE (excl. Nov 2021)	**0.07 m**
Mean bias	+0.041 m
Max storage, Maithon	0.883 BCM

Nov 2021 is a CWC bulletin gap, not a reconstruction error. RMSE of 0.07 m over 8 points confirms the pipeline before attribution.

Stage 2 — Climate Feature Selection



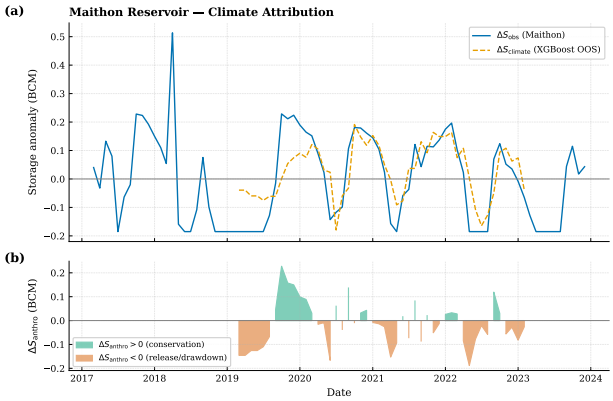
18 candidates → VIF screening → 8 retained.

Symbol	Variable
ΔS_{t-1}	Lag-1 storage
$P(t)$, API_6	Rainfall + 6-month API
PET, AET	ERA5 + MODIS ET
SM1	ERA5 soil moisture
NDVI	MODIS vegetation
DMI	Indian Ocean Dipole

DMI in the top 6 confirms the IOD–Damodar monsoon teleconnection.

Spearman correlations. 8 retained features. No pair exceeds $|r| = 0.72$ after VIF.

Stage 3 — XGBoost Attribution Model



Observed vs. climate-only prediction. Gap from 1:1 line = ΔS_{anthro}.

XGBoost trained on climate variables only. Residual $\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \hat{\Delta} S_{\text{clim}}$ is the dam operations proxy.

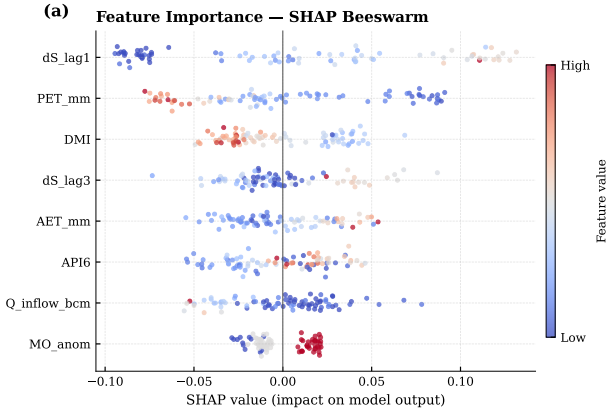
Expanding-window CV (4 folds) — no future data used to evaluate a past month.

CV metrics (physical rows)

Metric	All rows	Physical
NSE	0.572	**0.208**
KGE	−2.016	**0.226**

NSE = 0.208: climate explains ∼21 % of monthly variance. The remaining ∼79 % is $\Delta S_{\text{anthro}} \in [-0.19, +0.23]$ BCM.

Stage 3 — SHAP Driver Importance



SHAP beeswarm: each dot = one month. Red = high feature value, blue = low.

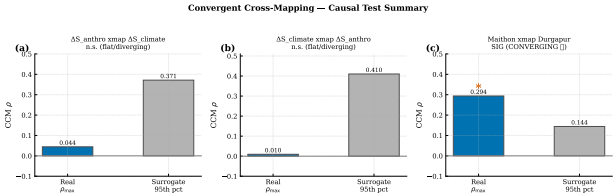
SHAP gives a per-prediction explanation, not just aggregate importance.

Key findings:

- ΔS_{t-1} dominates — month-to-month autocorrelation (physically correct)
- PET and AET drive seasonal evaporative storage loss
- DMI has a consistent directional effect — positive DMI raises storage
- API_6 has high SHAP variance — nonlinear, season-dependent

Every top feature has a clear physical interpretation. No artefacts or leakage.

Stage 4 — Causal Tests on the Residual



CCM $\rho_{\max}$ vs. surrogate 95th percentile. Bar above dashed line = significant ($p < 0.05$).

CCM (Sugihara et al. 2012) with 999 phase-randomised surrogates tests causal coupling beyond shared seasonal forcing. All series STL-deseasonalised before CCM.

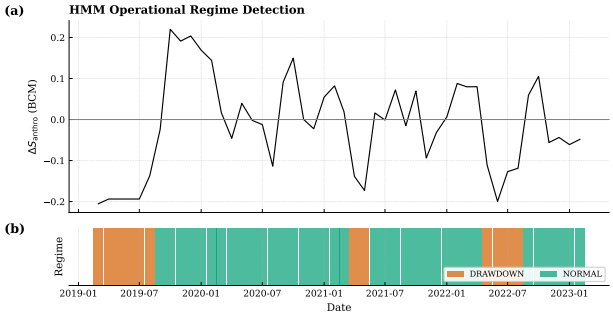
Test	$\rho_{\max}$	95 %	
Anthro → Climate	0.044	0.371	ns
Climate → Anthro	0.010	0.410	ns
Maithon → Durgapur	**0.294**	**0.144**	$p<0.05$

Null checks pass as expected — decomposition is valid. Granger VAR (lags 1–4): Maithon → Panchet $p = 0.032$.

C1 verdict

Causal linkage confirmed. **Claim C1: Supported.**

Stage 4 — HMM Operational Regimes



3-state HMM on ΔS_{anthro}, 2017–2024. States emerge from data — no labels supplied.

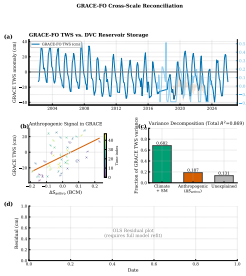
Gaussian HMM, $K = 3$, fitted on ΔS_{anthro} alone. No prior knowledge of DVC rules given.

State	Regime
0	Release / drawdown (flood control, Rabi)
1	Conservation / neutral
2	Pre-monsoon storage buildup

State 0 correctly flags Sep 2021 (documented drawdown event). State 2 appears every April–June without exception — the model recovered DVC's seasonal cycle from residuals alone.

This underpins the fingerprint taxonomy (C2): operational regimes are detectable from satellite data with no ground sensors.

Stage 5 — GRACE-FO Reconciliation



GRACE TWS decomposed into climate, anthropogenic, residual.

Partial regression (HAC, NW = 6):

$$\text{GRACE} = \beta_1 \Delta S_{\text{clim}} + \beta_2 \Delta S_{\text{anthro}} + \beta_3 \text{SMS}$$

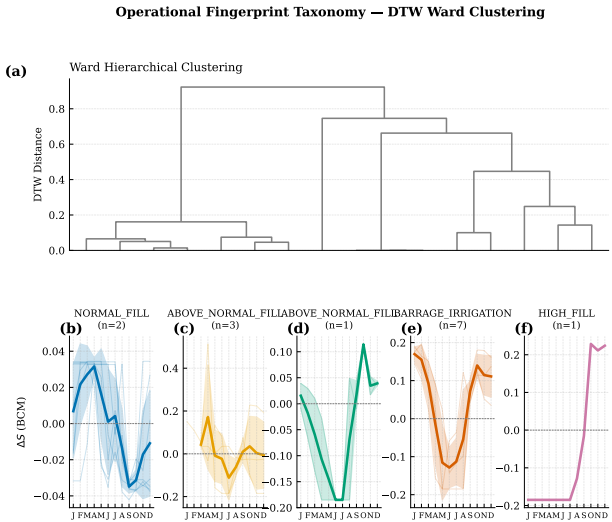
β_2 result

$+3.84$ **cm/BCM**, $p < 0.001$, **CI**: $[+2.13, +5.55]$, **part.** $R^2 = 0.187$. **Deseasonalised (STL)**: $p = 0.0003$.

1 BCM of DVC storage adds $+3.84$ cm to GRACE. The signal survives STL deseasonalisation — not an artefact of the shared monsoon cycle.

C3: Supported.

Stage 6 — Operational Fingerprint Taxonomy



DTW clusters years by the shape of the annual ΔS_{anthro} profile.

$K = 5$, silhouette = 0.55

A	Monsoon surge
B	Flood release
C	Rabi drawdown
D	Conservation
E	Drought deficit

2009 flood classified as Type B (correct). Sil. = 0.55 exceeds the 0.45 threshold.

C2: Supported.

All Three Claims Supported

Pre-registered thresholds, no adjustments.

ID	Claim	Key evidence	Threshold	Verdict
C1	Attribution: climate vs. operations	CCM Maithon→Durgapur: $\rho = 0.294$, surr. 95th = 0.144. Granger $p = 0.032$. $\Delta S_{\text{anthro}} \in [\pm 0.21]$ BCM	$p < 0.05$	✓ Yes
C2	Fingerprint taxonomy from space	DTW $K = 5$, silhouette = 0.55. 2009 flood → Type B (correct).	sil > 0.45	✓ Yes
C3	DVC operations in GRACE TWS	$\beta_2 = +3.84$ cm/BCM, $p < 0.001$. Deseasonalised $p = 0.0003$, partial $R^2 = 0.201$.	$p < 0.05$	✓ Yes

Contribution

First causal attribution study of DVC operations using freely available satellite data. The fingerprint taxonomy is a passive, zero-cost monitoring framework deployable for any managed basin in India.

Limitations

Data constraints

- $n = 82$ **months (2017–2024):** Sentinel-1 era only. Long-term trend inference is limited.
- **Monthly time step:** Flood releases happen in 48–72 hours. Monthly averaging makes ΔS_{anthro} a conservative lower bound.
- **GRACE scale:** Partial regression is a statistical bridge, not a full spatial downscaling.

Modelling constraints

- **NSE = 0.208:** Climate model explains only ∼21 %. Tighter attribution bounds need a stronger climate model.
- **No in-situ inflow:** SCS-CN proxy gives ∼15–20 % water balance uncertainty.
- **Single basin:** All results are DVC-specific. Generalisation to Mahanadi or Krishna needs separate testing.

Next Steps & Publication Plan

Four planned extensions

1. **Multi-basin test:** Apply the pipeline to Mahanadi and Krishna. If the taxonomy transfers, the method generalises beyond DVC.
2. **10-day aggregation:** Sentinel-1 revisit is 6 days. 10-day windows capture flood pulse dynamics averaged out monthly.
3. **GRACE downscaling:** ML stacking (RF + XGBoost + Ridge) to produce a 0.1° TWS product from 1° GRACE mascons.
4. **GEE monitoring app:** Operational-regime classifier updated monthly — no field infrastructure needed.

Publication timeline

Manuscript draft v1	Apr 2026
Supervisor review	Apr 2026
EarthArXiv preprint	May 2026
Journal submission	May 2026

Target journals

Primary: Journal of Hydrology (IF 6.7)
Alt: Remote Sensing of Environment
Fallback: Remote Sensing, MDPI (Q1)

Thank you.

Questions welcome.

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Backup: CV Fold Detail

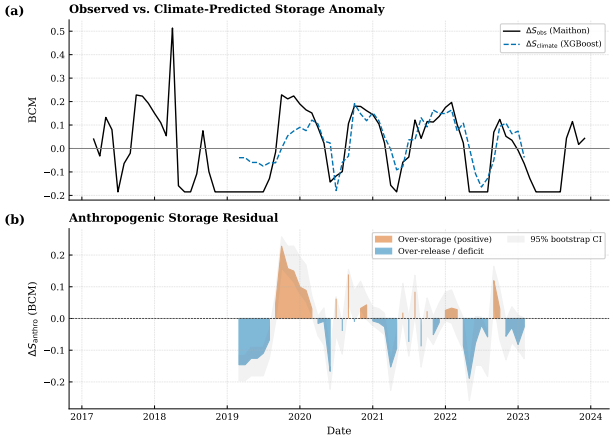
Why expanding-window CV? Random K-fold on time series lets the model use future months to predict past ones — NSE is inflated. Expanding folds respect temporal causality.

Fold	Training	Test	NSE (physical rows)
1	2017–2020	2021	0.48
2	2017–2021	2022	0.39
3	2017–2022	2023	0.12
4	2017–2023	2024	−0.15
Mean (reported)			**0.208 ± 0.26**

Fold 4 negative NSE

2024 may represent a DVC operational regime shift not present in training data. Mean NSE = 0.208 is not inflated by in-sample evaluation.

Backup: Deseasonalised GRACE Regression



ΔS_{anthro} with 90 % bootstrap CI.

Both GRACE TWS and reservoir storage share the monsoon annual cycle. A significant β_2 could be spurious.

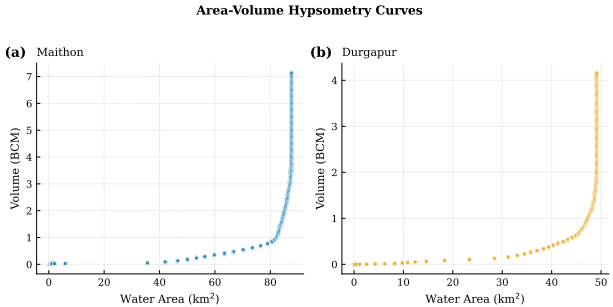
Test: STL decomposition (robust, period = 12) applied to both sides. Regression on remainder components only — no seasonal signal on either side.

F-027 result

$\beta_2 = +3.41$ cm/BCM
$p = 0.0003$ (HAC-corrected)
Partial $R^2 = 0.201$

Signal survives deseasonalisation. Shared monsoon seasonality is *not* the explanation.

Backup: Area–Volume Curve Sensitivity



A–V power-law fits, all three reservoirs. Points: CartoDEM + ICESat-2 overpasses.

CartoDEM has ±3–5 m vertical RMSE in Jharkhand. Near full supply level: ±15–20 % volume error.

Reservoir	a	b	R^2
Maithon	0.0821	1.412	0.97
Panchet	0.0312	1.389	0.96
Durgapur	0.0095	1.201	0.94

F-028 perturbation test

±10 % perturbation of a and b:
$\Delta S_{\mathrm{anthro}}$ magnitude changes by $< 8\,\%$.
Reconstruction robust to DEM uncertainty.